Hypothesis Testing

1. In statistics, **population** refers to the complete set of individuals or observations that we are interested in studying. However, in most real-world situations, it is not practical to study the entire population due to time, cost, or feasibility constraints. Therefore, we select a **sample**, which is a smaller subset of the population, to draw conclusions about the whole population.

2. A **parameter** is a numerical characteristic of a population, such as the population mean, variance or proportion, and it is usually unknown. On the other hand, a **statistic** is a numerical value calculated from the sample data, and it is used to estimate the corresponding population parameter. Thus, statistical analysis involves using sample statistics to make inferences about population parameters.

3. To make such inferences, we use the concept of a hypothesis, which is a statement or assumption about a population parameter. In hypothesis testing, we formulate two competing statements.

   - The first is the **null hypothesis (H_0)**, which represents the default assumption, typically stating that there is no effect, no difference, or no relationship between variables.
   - The second is the **alternative hypothesis (H_1)**, which contradicts the null hypothesis and represents the claim we want to investigate.

4. While making decisions based on sample data, there is always a possibility of error.

   - A **Type I error** occurs when we reject the null hypothesis even though it is actually true. The probability of committing this error is denoted by the significance level, usually represented by α (commonly 0.05, 0.01).
   - A **Type II error** occurs when we fail to reject the null hypothesis even though it is false.

Chi-Square Test of Independence of Attributes
The Chi-Square Test of Independence is a statistical method used to determine whether two categorical variables are independent or associated within a population. It is commonly applied when data are organized in a contingency table showing the frequency distribution across multiple categories.
The null hypothesis H_0 states that the two attributes are independent, while the alternative hypothesis H_1 states that there is a significant association between them.
The test statistic is computed as:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

where

- O_{ij} is the observed frequency in the i-th row and j-th column,
- E_{ij} is the expected frequency, given by: $E_{ij} = \dfrac{(\text{row total}) \times (\text{column total})}{\text{grand total}}$.

The test statistic follows a Chi-Square distribution with degrees of freedom: $\text{df} = (r-1)(c-1)$, where r is the number of rows and c is the number of columns in the table.

The decision is based on comparing the calculated χ^2 value with the critical value from the Chi-Square distribution table at a significance level α. If the test statistic exceeds the critical value, we reject H_0, concluding that the variables are not independent.

Example

1. Two sample polls of votes for two candidates A and B for a public office are taken, one from among the residents of rural areas and one from urban areas. The results are given in the side table. Examine whether the nature of the area is related to voting preference in this election. using the Chi-Square test of independence at the 5% level of significance. Given: The critical value of χ^2 at 5% level of significance for 1 degree of freedom is 3.84.

Area	Votes for		Total
	A	B	
Rural	620	380	1000
Urban	550	450	1000
Total	1170	830	2000

2. Out of 8000 graduates in a town 800 are females, out of 1600 graduate employees 120 are females. Use χ^2 to determine if any distinction is made in appointment on the basis of sex. Value of χ^2 at 5% level for one degree of freedom is 3.84.

3. A sociologist wants to study whether the type of employment sector (Government, Private, or Self-employed) is associated with an individual's preference for remote work. A survey of 300 people is conducted and the responses are tabulated. Test at the 5% level of significance whether there is a significant association between employment type and remote work preference. The critical value of χ^2 at 5% level of significance for 2 degrees of freedom is 5.99

Employment Type	Prefer Remote Work	Do Not Prefer Remote Work	Total
Government	40	60	100
Private	70	30	100
Self-employed	50	50	100
Total	160	140	300

4. A survey is conducted among 600 working individuals to examine whether type of employment is associated with gender. The results are summarized in the table. Test whether the type of employment is independent of gender using the Chi-Square test of independence at the 5% and 10% levels of significance.
Given: The critical value of χ^2 at 5% level of significance for 2 degrees of freedom is 5.99. The critical value of χ^2 at 10% level of significance for 2 degrees of freedom is 4.61

Gender	Government	Private	Self-Employed	Total
Male	120	150	90	360
Female	80	100	60	240
Total	200	250	150	600

5. Below is given the distribution of hair colours for either sex in a university. Test the homogeneity of hair colour for boys and girls using the Chi-Square test at the 1%, 5%, and 10% levels of significance.
Given: The critical value of χ^2 at 1% level of significance for 4 degrees of freedom is 13.28. The critical value of χ^2 at 5% level of significance for 4 degrees of freedom is 9.49. The critical value of χ^2 at 10% level of significance for 4 degrees of freedom is 7.78

Hair colour	Fair	Red	Medium	Dark	Jet black	Total
Boys	592	119	849	504	36	2100
Girls	544	97	677	451	14	1783
Total	1136	216	1526	955	50	3883

6. A survey is conducted among 300 individuals to study whether the level of education is associated with their preferred mode of transportation. The results are summarized in the following table:

Education Level	Bus	Car	Bike	Total
School	40	30	30	100
Graduate	30	50	20	100
Postgraduate	20	40	40	100
Total	90	120	90	300

Test whether the level of education is independent of the preferred mode of transportation using the Chi-Square test of independence at the 5% and 10% levels of significance.

Given: The critical value of χ^2 at 5% level of significance for 4 degrees of freedom is 9.49. The critical value of χ^2 at 10% level of significance for 4 degrees of freedom is 7.78.